MALWANDLA HLONGWANE

Junior Database Developer | SQL Developer

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PROFESSIONAL SUMMARY

Final-year Diploma in ICT (Applications Development) student at the University of Mpumalanga with hands-on relational database work across five real applications. Designed a 26-table MySQL schema with 39 foreign keys and 76 indexes, built a tamper-evident audit log in PostgreSQL, and used transactions and row locks to prevent double bookings. Skilled in ER modelling, normalisation, SQL, indexing, data security and connecting databases to REST APIs. Seeking a Junior Database Developer, SQL Developer, Junior DBA or data-focused graduate role.

TECHNICAL SKILLS

- **Database Systems:** MySQL, PostgreSQL, SQLite, Room (Android), MySQL Workbench, pgAdmin, phpMyAdmin
- **SQL:** DDL and DML, complex joins, subqueries, set operators, aggregate and analytical functions, stored procedures, triggers
- **Design:** ER modelling, normalisation, conceptual, logical and physical data models, primary and foreign keys, constraints, indexing
- **Integrity & Performance:** transactions, row-level locking, concurrency control, query optimisation, schema migrations
- **Data Security:** prepared statements (SQL injection prevention), password hashing, AES-256 encryption at rest, audit logging, role-based access
- **Application Integration:** PDO, JDBC, JPA/Hibernate, mysql2 (Node.js), REST APIs, CSV export; Java, PHP, JavaScript
- **Also covered in coursework:** distributed databases, data warehousing, star schemas, business intelligence, Big Data and NoSQL (MongoDB, Neo4j, Hadoop)

PROJECT EXPERIENCE

Taskify | Campus Task & Equipment Marketplace

Full-stack web application | MySQL database design
Tech: Node.js, Express, MySQL, JavaScript, HTML, CSS, REST API, JWT, Render

- Designed a 26-table MySQL schema with 39 foreign keys and 76 indexes supporting users, tasks, sales, rentals, disputes and payments.
- Prevented duplicate bookings and payments under concurrent use with row-level locks and conditional updates.
- Combined 28 database repair scripts into one ordered runner and regenerated the schema file, verified against the live database (325 columns) with no rows lost.
- Covered database-driven workflows with 134 automated tests across 13 suites, all passing.

GitHub: github.com/Malwandla21-ict/taskify | **Live demo:** taskify-frontend-ol07.onrender.com

StokVault | Stokvel Contribution & Payout Platform

Enterprise Java | Group project (DICT312) | PostgreSQL
Tech: Java 21, Jakarta EE 11 (Faces, REST, EJB, CDI, JPA, JMS, Batch), Payara Server, PostgreSQL, Maven, Docker, GitHub Actions, JUnit 5

- Modelled contributions, payout cycles and per-group member roles in PostgreSQL using JPA entities.
- Built a SHA-256 hash-chained audit log, with a PostgreSQL trigger that blocks UPDATE, DELETE and TRUNCATE on audit records.
- Encrypted ID numbers at rest with AES-256-GCM and enforced uniqueness with an HMAC.

GitHub: github.com/Malwandla21-ict/StokVault

UMP Events | Campus Event Booking Website

PHP web application | Started as a 2nd-year group project, since rebuilt
Tech: PHP 8.2, MySQL, PDO, MVC, HTML, CSS, JavaScript, Apache (XAMPP)

- Moved all SQL into one repository class per table, using PDO prepared statements to prevent SQL injection.
- Prevented overbooking with transactions, row locks and a unique constraint on bookings.

- Built admin reports (fill rates, recent bookings) and CSV attendee exports protected against formula injection.

GitHub: github.com/Malwandla21-ict/Ump-Event-Booking-web

ADDITIONAL PROJECTS

HealthWise | AI Symptom Checker Android App

Native Android application

Tech: Java, Android SDK, XML, Material Components 3, Room (SQLite), Retrofit, Gson, AlarmManager, Hugging Face API

- Designed Room (SQLite) tables for check history, pill reminders, dose logs and health profile, with migrations that keep user data across schema versions.

GitHub: github.com/Malwandla21-ict/HealthWise

FUEL SA | Fuel Ordering & Station Management System

Java desktop application

Tech: Java 17, JavaFX 21, FXML, CSS, SQLite (JDBC), Maven, BulkSMS REST API

- Built an SQLite database with DAO classes over JDBC for customers, vehicles and fuel requests, feeding revenue and top-customer reports.

GitHub: github.com/Malwandla21-ict/FuelSA

EDUCATION

Diploma in Information and Communication Technology (Applications Development)

University of Mpumalanga | Expected completion: 2026 | Academic average: 75–78%

Relevant coursework:

- **Information Systems III (Database Systems):** ER and advanced data modelling, normalisation, SQL programming, join and set operators, database architectures
- **Information Systems III (Advanced Database Concepts):** transaction management, concurrency control, performance tuning, distributed DBMS, database security, Big Data, NoSQL, BI and data warehousing
- **Web Development:** MySQL with PHP (PDO), multi-table queries, foreign keys and joins